Semester-IV

Course Code- 119401 Transportation Engineering **3 0 0 3**

Unit- 1.0 **8 hrs**
Highway development and planning-Classification of roads, road development in India, Current road projects in India; highway alignment and project preparation.

Unit- 2.0 **5 hrs**
Geometric design of highways-: Introduction; highway cross section elements; sight distance, design of horizontal alignment; design of vertical alignment; design of intersections, problems.

Unit- 3.0 **6 hrs**
Traffic engineering & control- Traffic Characteristics, traffic engineering studies, traffic flow and capacity.

Unit- 4.0 **8 hrs**
Traffic regulation and control; design of road intersections; design of parking facilities; highway lighting; problems.

Unit- 5.0 **5 hrs**
Pavement materials- Materials used in Highway Construction- Soils, Stone aggregates, bituminous binders, bituminous paving mixes; Portland cement and cement concrete: desirable properties, tests, requirements for different types of pavements. Problems.

Unit- 6.0 **10 hrs**
Design of pavements- Introduction; flexible pavements, factors affecting design and performance; stresses in flexible pavements; design of flexible pavements as per IRC; rigid pavements- components and functions; factors affecting design and performance of CC pavements; stresses in rigid pavements; design of concrete pavements as per IRC; problems.

Text/ Reference:-

1. Khanna, S.K., Justo, C.E.G and Veeraragavan, A, 'Highway Engineering', Revised 10th Edition, Nem Chand & Bros, 2017.
2. Kadiyalai, L.R., ' Traffic Engineering and Transport Planning', Khanna Publishers.
3. Partha Chakraborty, ' Principles of Transportation Engineering, PHI Learning.
4. Fred L. Mannering, Scott S. Washburn, Walter P. Kilareski,'Principles of Highway Engineering and Traffic Analysis', 4th Edition, John Wiley.
5. Srinivasa Kumar, R, Textbook of Highway Engineering, Universities Press, 2011.
6. Paul H. Wright and Karen K. Dixon, Highway Engineering, 7th Edition, Wiley Student Edition, 2009.